

Xiaoqing PAN

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Genome Biology and Epigenetics (GBE) division of the department of Biology, Utrecht University
Utrecht, Netherlands

Re: Postdoctoral position in gene regulation

Dear Dr. Vlaming,

I am writing to express my interest in the postdoc position in your group. Your lab's focus on deciphering the molecular role of RNA Polymerase II grab my interest. As a doctoral candidate in microbiology at Hans-Knöll-Institute in Jena, Germany (thesis submitted in 01.2026), my research mainly focused on double-stranded RNA (dsRNA) metabolism in infectious diseases.

My research background focused on host-pathogen interactions, specializing how human immune cells (PBMCs) respond to inflammatory triggers, where I mainly contributed to bioinformatic analysis. With a focus on A-to-I RNA modification and RNA splicing analysis, I learnt about dsRNA forming and immunology. With unpublished results, one hypothesis was that upon one pathogenic fungal stimulation, the elongation rate of transcription in PBMCs was slowed down, which resulted in alternative splicing with more introns retained in mature RNA. With this project, I have successfully published a first-author review in *ACS Infectious Diseases*.

Moreover, I was also studying the RNA interference (RNAi) pathway in the human pathogen *Aspergillus fumigatus*. By investigating sRNA landscape in *A. fumigatus*, we have found out that the fungus produced canonical sRNA under stress growth environments. In this project, I focused on analyzing sRNA-seq (bioinformatics), measuring fungal biomass (wet-lab) to dissect how RNAi machinery processes dsRNA and regulates fungal development, work that resulted in first-author and co-author in *Molecular Microbiology* and *RNA*.

Based on the job description, the prerequisites align with my background in molecular biology. I have knowledge in analyzing genome/transcriptome data though large-scale datasets analyzing experience is lacking. While my expertise lies on host-fungal system, the core principles of RNA Polymerase II function, RNA splicing machinery are broadly applicable across organisms.

What draws me particularly to your lab is the opportunity to dive into molecular biology in human cells. I studied RNA in both masters and PhD in different organisms. Through host-fungal study, I've learnt human immunology. This position can provide me with a better opportunity to be an expert in gene regulation field, which can be applied across organisms. In the future, I hope to contribute to the medical industry, and working as part of this project would present a great stepping stone to gain broader expertise beyond fungal pathogens.

My PhD supervisor, Dr. Matthew Blango (matthew.blango@leibniz-hki.de) and Dr. Olaf Kniemeyer (olaf.kniemeyer@leibniz-hki.de) have agreed to serve as referees. Thank you for your time and consideration.

Best,

Xiaoqing

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SUMMARY

Doctoral candidate in **Microbiology** specializing in RNA biology in the context of host-pathogen interactions. Proficient in R, Linux and NGS data processing within HPC environments (Slurm). Experienced in RNA-seq analysis, prediction of A-to-I RNA editing and alternative splicing and microbial assays. Passionate about integrating bioinformatics to address questions in biotechnology and immunology. Published five papers and two in preparation.

EDUCATION

PhD | Microbiology, Friedrich Schiller University, Germany 2026
Thesis: RNA regulation in host-fungal interactions: investigation of the host pitranscriptome and fungal small RNAs

MSc | Bioengineering, Chinese Academy of Sciences, China 2021
Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively

PROFESSIONAL EXPERIENCES

Human PBMC transcriptome changes upon fungal stimulations

- Designing a high-throughput **Snakemake workflow** to quantify differential expression and alternative splicing events, characterizing the dynamic response of human PBMCs to fungal pathogens.
- Leading a six people-involved project. Collaborate with Fritz-Lipmann-Institut, Germany
- DOI: [10.1021/acsinfecdis.4c00598](https://doi.org/10.1021/acsinfecdis.4c00598)

RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Develop both sRNA- and mRNA-seq pipeline for differential expression gene analysis.
- Experimental validation support for determining the **role of RNAi machinery and sRNA** using Northern Blot, PCR, qPCR assays.
- Leading and involving a twenty people project. Collaborate with The University of Manchester, the University of Jena
- DOI: <https://doi.org/10.1111/mmi.70042> and DOI: [10.1261/rna.079350.122](https://doi.org/10.1261/rna.079350.122)

Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analysis of human extracellular small RNA transcriptome.
- Involved in a ten people project. Collaborate with the University of Jena.

Non-coding RNA analysis of rice-*Xoo* interaction

- Experimental validation support for determining **non-coding RNAs (sRNAs, lncRNAs, circRNAs) in rice responding to fungal infections** using PCR, qPCR, LC-MS, assays.
- Leading a five people-involved project. Collaborate with Sun Yat-Sen University, China
- DOI: [10.1186/s42483-023-00188-8](https://doi.org/10.1186/s42483-023-00188-8) and DOI: [10.1016/j.jplph.2022.153905](https://doi.org/10.1016/j.jplph.2022.153905)

Teaching

- Microbiology and Molecular Biology - MMB005, 2022 - 2025
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TECHNICAL SKILLS

- **Bioinformatics:** HPC, Snakemake, Genomics, RNA-seq, SHAPE-seq, Alternative splicing, Data visualization
 - **Molecular Biology:** DNA/RNA/protein isolation, Genetic Engineering, Northern Blot, qPCR, PCR, LC-MS, Gibson Assembly Cloning, Cultivation of microorganisms
 - **Programming & Tools:** R, python, Bash, Git, Docker, Linux
 - **Language:** English (fluent), German (B1)
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PUBLICATIONS

Pan X*, Kelani AA*, *et al.* (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. *Mol Microb* 1-12.

Pan X*, *et al.* (2024). The past, present, and future of RNA modifications in infectious disease research. *ACS Infect Dis* 10(12), 4017-4029.

Xia K*, **Pan X***, *et al.* (2023). X00-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of *OsARAB* in rice. *Phytopathol Res* 5, 22.

Xia K*, **Pan X***, *et al.* (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3*, *OsNPF2.4*, and *OsAGO1a*, respectively. *J. Plant Physiol* 208, 153905.

Kelani AA, ..., **Pan X**, ..., Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. *RNA* 29(7), 1033-1050.

CONFERENCES

2025 - Oral presentation: German speaking Mycological Society (DMykG), Germany

2025 - Poster: Gordon Research Seminar - RNA Editing, Italy

2024 - Oral presentation: Jena RNA Club Symposium, Germany

2024 - Poster: RNA Society Meeting, UK // Poster

2023 - Poster: EMBL - The Non-coding Genome Symposium, Germany

AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023